

EXPERIMENT-1

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Course code: MLA0305

Course Name: REINFORCEMENT LEARNING

Slot: B

1.Installation and Configuration of Anaconda Navigator, Python, TensorFlow, Keras and Gymnasium

Aim

To install and configure **Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium** and verify that all required tools are working correctly.

Procedure

1. Install and open **Anaconda Navigator**.
2. Install **Python** and configure the Python environment.
3. Open **Anaconda Prompt/Command Prompt**.
4. Install TensorFlow, Keras, and Gymnasium using pip.
5. Open **Jupyter Notebook** through Anaconda Navigator.
6. Import the installed libraries and verify their versions.

Python code:

```
import sys
```

```
import tensorflow as tf
```

```
import keras
```

```
import gymnasium as gym
```

```
print("Python Version :", sys.version.split()[0])
```

```
print("TensorFlow Version :", tf.__version__)
```

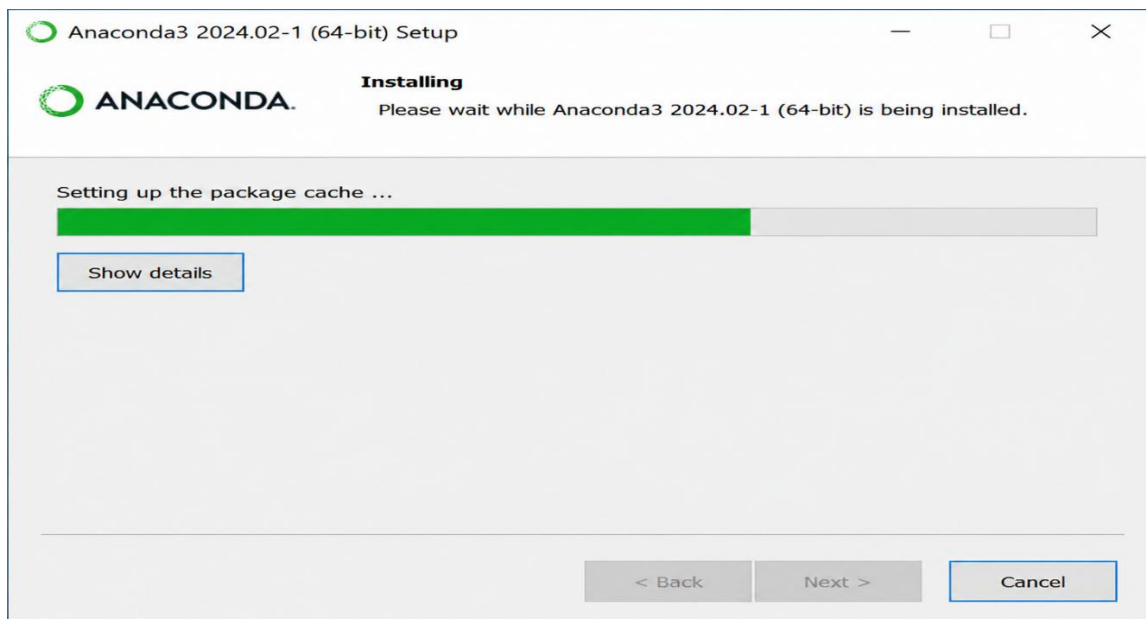
```
print("Keras Version   :", keras.__version__)
```

```
print("Gymnasium Version :", gym.__version__)
```

```
print("\nAll packages installed and imported successfully.")
```

Output:

Anaconda installation:



Python installation:



Tensorflow installation:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\akhil>pip install tensorflow
Collecting tensorflow
  Downloading tensorflow-2.16.1-cp310-cp310-win_amd64.whl (589.9 MB)
    _____ 589.9/589.9 MB 6.2 MB/s eta 0:00:00
Collecting absl-py>=1.0.0 (from tensorflow)
  Downloading absl_py-2.1.0-py3-none-any.whl (133 kB)
    _____ 133.1/133.1 kB 7.6 MB/s eta 0:00:00
Collecting astunparse>=1.6.0 (from tensorflow)
  Downloading astunparse-1.6.3-py2.py3-none-any.whl (12 kB)
Collecting flatbuffers>=23.5.26 (from tensorflow)
  Downloading flatbuffers-24.3.25-py2.py3-none-any.whl (26 kB)
Collecting gast<=0.4.0,>=0.2.1 (from tensorflow)
  Downloading gast-0.4.0-py3-none-any.whl (9.8 kB)
Collecting google-pasta>=0.1.1 (from tensorflow)
  Downloading google_pasta-0.2.0-py3-none-any.whl (57 kB)
    _____ 57.5/57.5 kB 3.2 MB/s eta 0:00:00
Collecting h5py>=3.10.0 (from tensorflow)
  Downloading h5py-3.11.0-cp310-cp310-win_amd64.whl (2.7 MB)
    _____ 2.7/2.7 MB 6.1 MB/s eta 0:00:00
...
Installing collected packages: absl-py, astunparse, flatbuffers, gast, google-pasta, h5py, . . . ,
tensorflow
Successfully installed absl-py-2.1.0 astunparse-1.6.3 flatbuffers-24.3.25 gast-0.4.0 google-pasta-
-0.2.0 h5py-3.11.0 .... tensorflow-2.16.1

C:\Users\akhil>
```

Keras:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\akhil>pip install keras
Collecting keras
  Downloading keras-3.3.3-py3-none-any.whl (1.2 MB)
    _____ 1.2/1.2 MB 6.7 MB/s eta 0:00:00
Collecting absl-py (from keras)
  Using cached absl_py-2.1.0-py3-none-any.whl (133 kB)
Collecting numpy (from keras)
  Using cached numpy-1.26.4-cp310-cp310-win_amd64.whl (15.8 MB)
Collecting rich (from keras)
  Downloading rich-13.7.1-py3-none-any.whl (240 kB)
    _____ 240.7/240.7 kB 3.8 MB/s eta 0:00:00
Collecting namex (from keras)
  Downloading namex-0.0.8-py3-none-any.whl (5.9 kB)
Collecting h5py (from keras)
  Using cached h5py-3.11.0-cp310-cp310-win_amd64.whl (2.7 MB)
Collecting optree (from keras)
  Downloading optree-0.12.1-cp310-cp310-win_amd64.whl (269 kB)
    _____ 269.1/269.1 kB 5.1 MB/s eta 0:00:00
Installing collected packages: namex, numpy, h5py, absl-py, optree, rich, keras
```

Gymnasium Installation:

```
Successfully installed absl-py-2.1.0 h5py-3.11.0 keras-3.3.3 namex-0.0.8 numpy-1.26.4 optree-0.12.1 rich-7.1
```

```
C:\Users\akhil>
```

```
C:\Windows\System32\cmd.exe
```

```
Microsoft Windows [Version 10.0.22631.3155]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\akhil>pip install gymnasium
```

```
Collecting gymnasium
```

```
  Downloading gymnasium-0.29.1-py3-none-any.whl (953 kB)
```

```
953.9/953.9 kB 3.7 MB/s eta 0:00:00
```

```
Collecting numpy>=1.21.0 (from gymnasium)
```

```
  Using cached numpy-1.26.4-cp310-cp310-win_amd64.whl (15.8 MB)
```

```
Collecting cloudpickle>=1.2.0 (from gymnasium)
```

```
  Using cached cloudpickle-3.0.0-py3-none-any.whl (20 kB)
```

```
Collecting typing-extensions>=4.3.0 (from gymnasium)
```

```
  Using cached typing_extensions-4.11.0-py3-none-any.whl (34 kB)
```

```
Collecting farama-notifications>=0.0.1 (from gymnasium)
```

```
  Downloading Farama_Notifications-0.0.4-py3-none-any.whl (2.5 kB)
```

```
Installing collected packages: farama-notifications, typing-extensions, numpy, cloudpickle, gymnasium
```

```
Successfully installed cloudpickle-3.0.0 farama-notifications-0.0.4 gymnasium-0.29.1 numpy-1.26.4 typing-extensions-4.11.
```

```
C:\Users\akhil>
```

Result

Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium were successfully installed and configured. The installed packages were imported successfully and their versions were verified.